

Federated Schedule

Behavioral-Aware System Heterogeneity
in Federated Learning

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COMP 5970 · Federated Learning

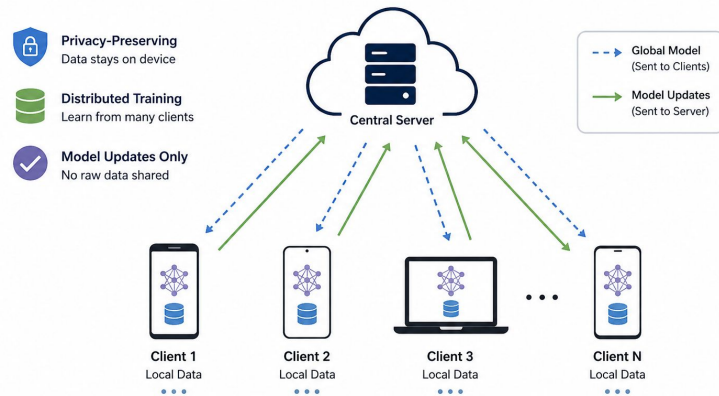


Federated Learning Overview

- "Bring the model to the data"
- Privacy-preserving — data stays on device
- Distributed clients coordinate without sharing raw data

Federated Learning Architecture

"Bring the model to the data, not the data to the server."



The Behavioral Gap



Hardware

Plugged vs. Unplugged



Humans

Unpredictable behavior



Time $\neq$ Reliability

Availability not guaranteed



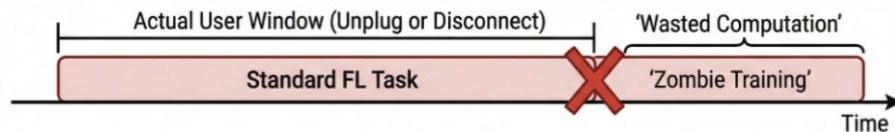
Standard FL assumes time availability implies reliable participation — this is false.

The Zombie Training Problem

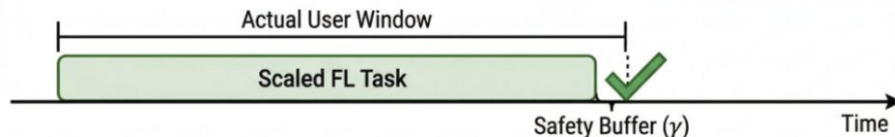
- Training interrupted before completion
- Updates are discarded — no contribution to global model
- Wasted compute + bandwidth on every failed round

"Before vs. After" Comparison: The Zombie Training Problem vs. Federated Schedule Solution

A. Scenario A: Zombie Training – The Problem



B. Scenario B: Federated Schedule – The Solution



Key Idea — Proactive Scheduling

FedAvg / FedProx

REACTIVE

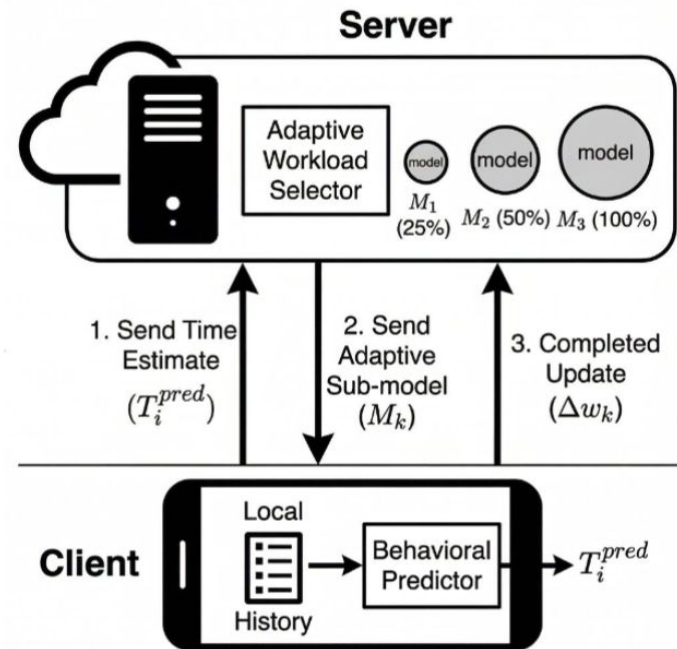
Schedule clients that respond; accept dropouts as inevitable

Federated Schedule

PROACTIVE

Predict availability before assigning work; prevent zombie training

Predict availability before training begins.



Charging Window Prediction

- Predict usable training time per device
- Uses local device history (charging patterns)
- Privacy preserved — no raw data leaves the device

$$T_i^{pred} = E[\text{remaining charging time} \mid \text{history}]$$

Expectation conditioned on observed device history

Adaptive Workload Selection

- Match workload to predicted time
- Long window → more training epochs
- Short window → fewer epochs
- Safety buffer γ prevents overcommitment

$$T_i^{usable} = \gamma \cdot T_i^{pred}$$

$\gamma \in (0, 1)$ — safety factor

Reliability-Weighted Aggregation

- Weight model updates by device reliability
- Stable, long-window clients contribute more
- Reduces influence of unreliable participants

$$\alpha_i = \frac{n_i \cdot \phi(T_i^{pred})}{\sum_j n_j \cdot \phi(T_j^{pred})}$$

n_i = local samples $\phi(\cdot)$ = reliability scoring function

Simulation Setup

1,000

Clients

200

Rounds

CIFAR-10

Dataset

Mixed

Behavioral Profiles

Baselines:

FedAvg

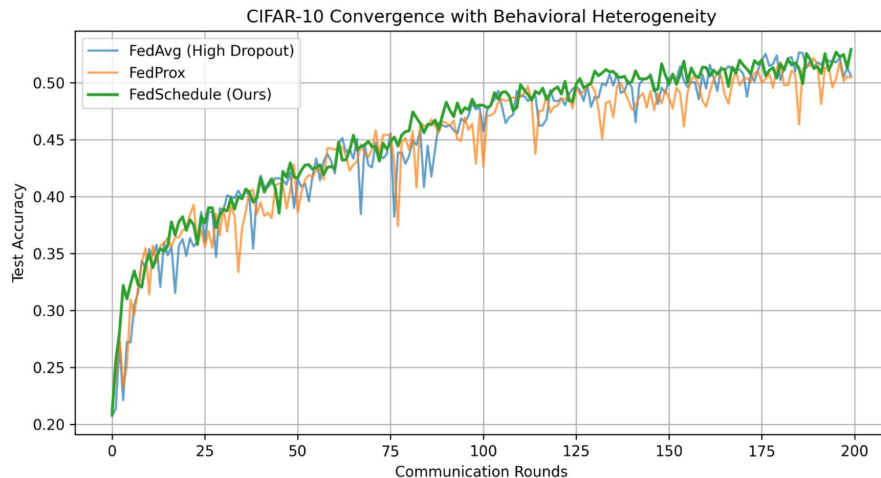
FedProx

Federated Schedule

Behavioral profiles: stable plugged-in, erratic, mobile — mixed heterogeneously across clients.

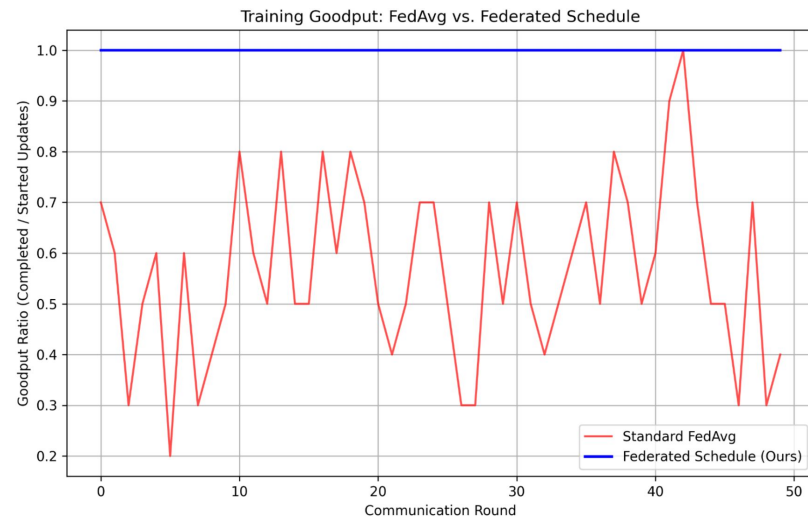
Accuracy Results

- Stable convergence under high client dropout rates
- Reduced model drift compared to reactive baselines



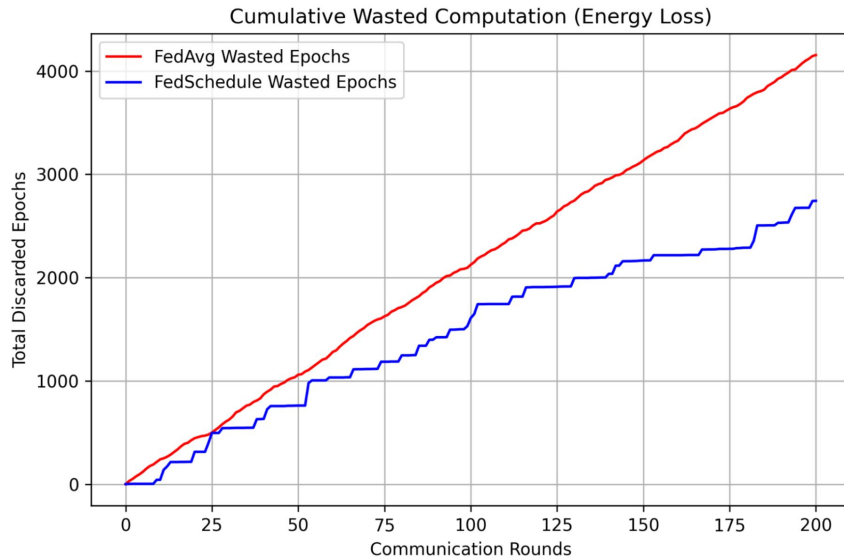
Goodput — Efficiency Metric

- $\text{Goodput} = \text{Completed updates} / \text{Started updates}$
- Measures actual system efficiency end-to-end
- Near-perfect adherence with Federated Schedule



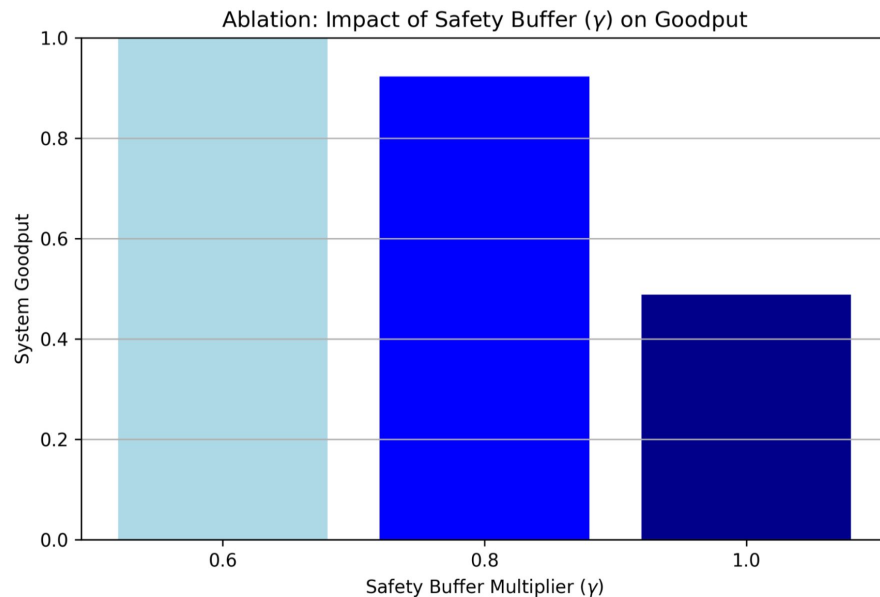
Wasted Computation

- Tracks discarded updates per round
- FedAvg wastes heavily under dropout conditions
- Federated Schedule substantially reduces waste



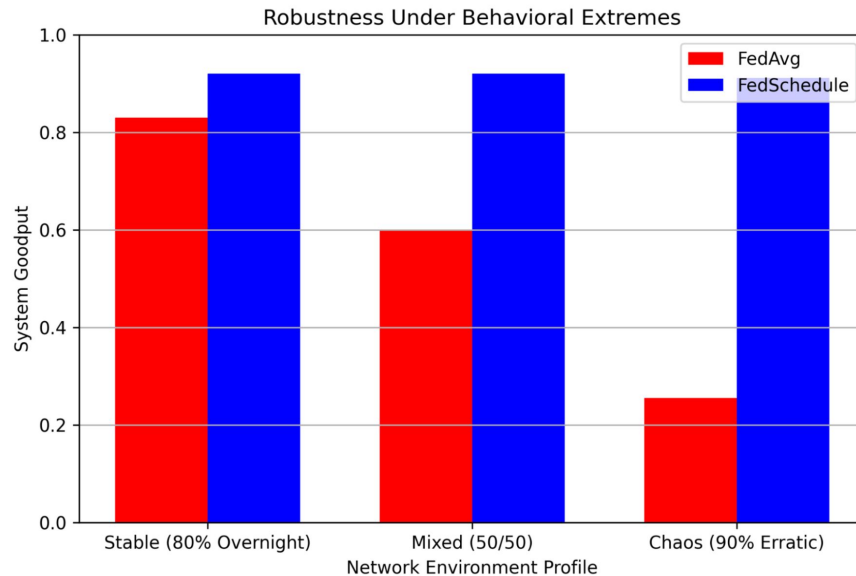
System Tuning — γ Ablation

- γ controls the safety vs. utilization tradeoff
- Low $\gamma \rightarrow$ safe but underutilizes training windows
- High $\gamma \rightarrow$ maximizes use, risks incomplete rounds
- Optimal $\gamma \approx 0.8$ across experiments



Robustness Under Chaos

- Stress test: 90% erratic / unpredictable users
- FedAvg degrades sharply — high dropout collapses convergence
- Federated Schedule maintains stable global training



Conclusion

- 1 Behavioral heterogeneity is a first-class problem in FL
- 2 Zombie training is preventable with proactive scheduling
- 3 Adaptive workloads improve utilization without sacrificing safety
- 4 More stable global training under realistic device conditions

Questions

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Federated Schedule — Behavioral-Aware System Heterogeneity in FL

